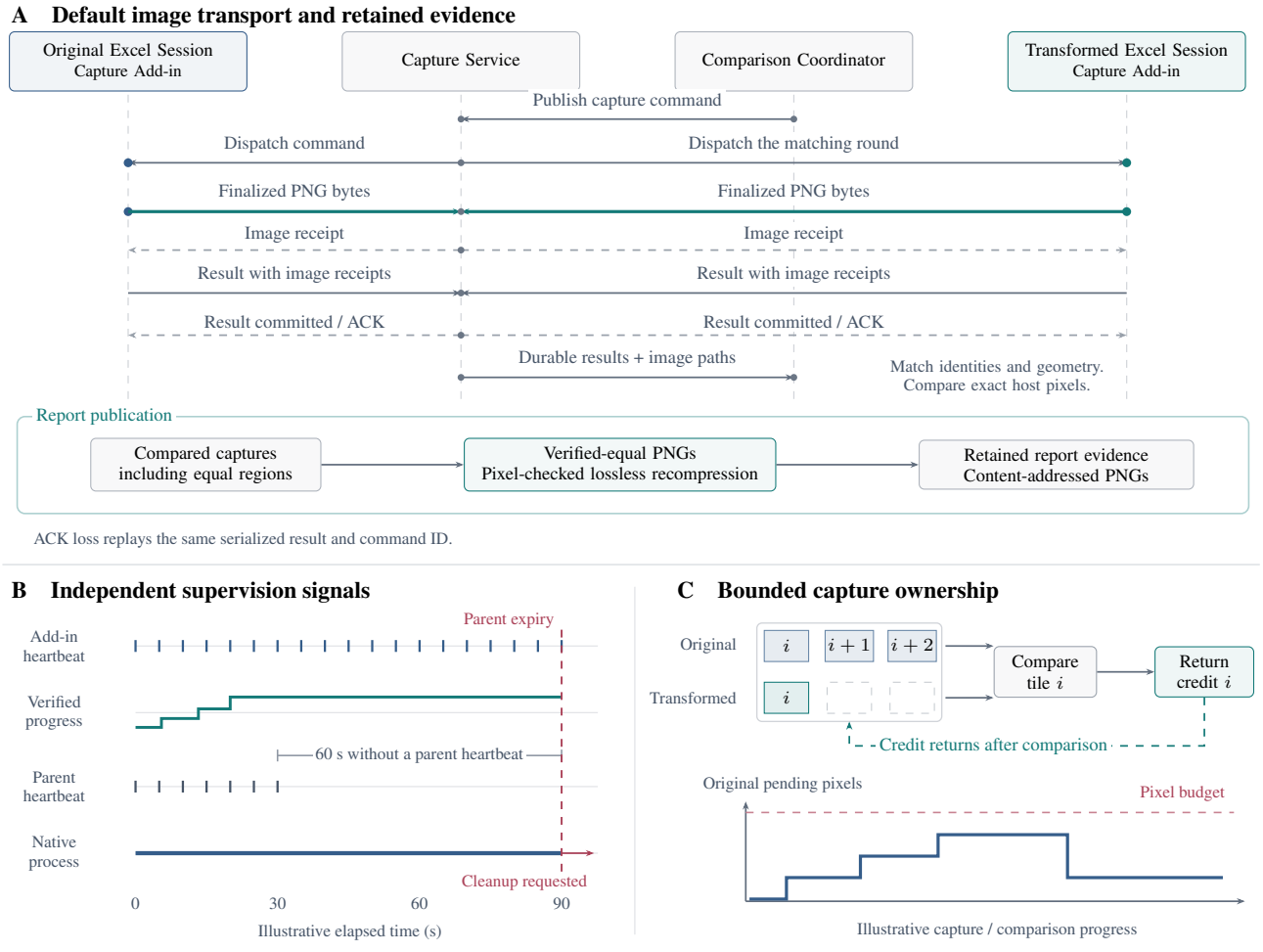
A Default image transport and retained evidence
Original Excel Session
Capture Add-in
Capture Service
Comparison Coordinator
Transformed Excel Session
Capture Add-in
Publish capture command
Dispatch command
Dispatch the matching round
Finalized PNG bytes
Finalized PNG bytes
Image receipt
Image receipt
Result with image receipts
Result with image receipts
Result committed / ACK
Result committed / ACK
Durable results + image paths
Match identities and geometry.
Compare exact host pixels.
Report publication
Compared captures
including equal regions
Verified-equal PNGs
Pixel-checked lossless recompression
Retained report evidence
Content-addressed PNGs
ACK loss replays the same serialized result and command ID.
B Independent supervision signals
Parent expiry
Add-in
heartbeat
Verified
progress
60 s without a parent heartbeat
Parent
heartbeat
Native
process
Cleanup requested
0
30
60
90
Illustrative elapsed time (s)
C Bounded capture ownership
Original
Transformed
i
i + 1
i + 2
i
Compare
tile i
Return
credit i
Credit returns after comparison
Original pending pixels
Pixel budget
Illustrative capture / comparison progress